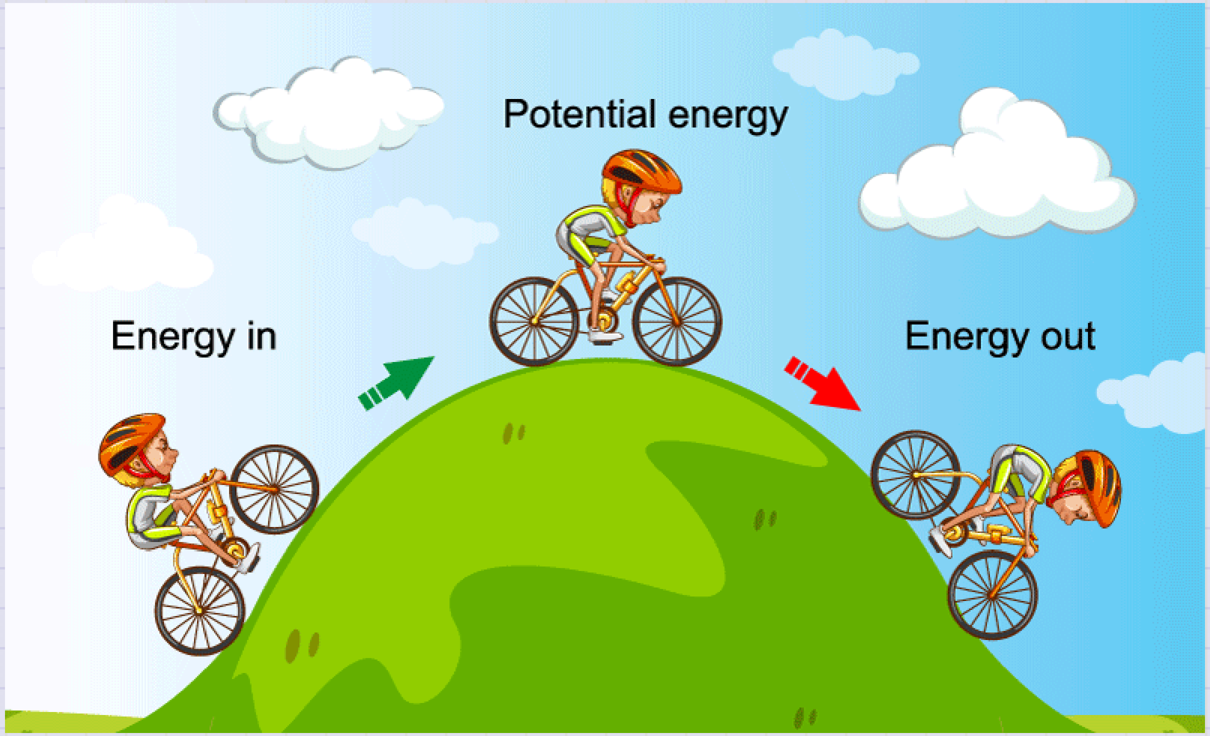
Day 12 - Conservation of Energy

$$\Delta E_{system} = W + Q$$



Announcements

- HW 4 is due Friday
- Midterm 1 will be available next Monday
  - Four exercises (3 physics; 1 project work)
  - You may work together; and ask any questions
  - DC will not work exercises from Midterm

Move Mihir's office hours?

1. Yes
2. No
3. Maybe?

Reminder of our Midterm Procedures

- The take-home midterms will be open for almost two weeks; you can often start some exercises early as they cover older material.
- They are meant to be challenging, but we will provide you with the resources and support you need to complete them.
- There is no homework due during the period in which the midterm is assigned.
- In contrast to homework assignments, DC will not work any exercises directly

You may work closely together with me, Mihir, and your classmates, but you and your partner must write up your own solutions.

Seminars this week

MONDAY, September 22, 2025

Condensed Matter Seminar 4:10 pm,1400 BPS, In Person and Zoom, Host ~ Philip Crowley

Speaker: Dean Lee, Michigan State University

Title: Algorithms for nuclear many-body systems

Zoom Link: https://msu.zoom.us/j/93613644939

Meeting ID: 936 1364 4939

Password: CMP

Seminars this week

WEDNESDAY, September 24, 2025

FRIB Nuclear Science Seminar, 11:00am., online via Zoom, (Zoom Only)

Speaker: Hongwei Zhao of IMPCAS

Title: High intensity heavy-ion accelerator facility and key technology R&D

Please see website for full abstract.

Please click the link below to join the webinar:

Join Zoom: https://msu.zoom.us/j/96657965451?pwd=Isaf23sK5agzaao0Kwaei7AaWHkc4W.1

Meeting ID: 966 5796 5451

Passcode: 479842

Seminars this week

WEDNESDAY, September 24, 2025

Astronomy Seminar, 1:30 pm, 1400 BPS, In Person and Zoom, Host~

Speaker: Kosuke Namekata, NAOJ

Title:

Zoom Link: https://msu.zoom.us/j/887295421?pwd=N1NFb0tVU29JL2FFSkk0cStpanR3UT09

Meeting ID: 887-295-421

Passcode: 002454

Seminars this week

THURSDAY, September 25, 2025

Colloquium, 3:30 pm, 1415 BPS, in person and zoom. Host ~

Refreshments and social half-hour in BPS 1400 starting at 3 pm

Speaker: Wolfgang Kerzendorf, MSU - (PTRC)

Title: Calibrating Stellar Explosions as Probes of the Evolving Universe

Background:

For more information and to schedule time with the speaker, see the colloquium calendar at https://pa.msu.edu/news-events-seminars/colloquium-schedule.aspx

Zoom Link: https://msu.zoom.us/j/94951062663

Password: 2002 Or complete link: https://msu.zoom.us/j/94951062663?pwd=c48uM25P9UsRVuR74rkOioOWgpoxgC.1

Seminars this week

FRIDAY, September 26, 2025

QuIC Seminar, 12:30pm, -1:30pm, 1300 BPS, In Person

Speaker: Alexei Bazavov, MSU

Title: Efficient State Preparation for the Schwinger Model

Full Scheule is at: https://sites.google.com/msu.edu/quic-seminar/

For more information, reach out to Ryan LaRose

Seminars this week

FRIDAY, September 26, 2025

IReNA Online Seminar, 2:00 pm, In Person and Zoom, FRIB 2025 Nuclear Conference Room, Light refreshments will be served at 1:50pm.

Hosted by: Steffen Turkat (TU Dresden, Germany)

Speaker: Dominik Koll, HZDR, Germany

Title: The search for freshly synthesized radionuclides from stellar explosions on Earth

Zoom Link: https://msu.zoom.us/j/827950260

Password: JINA

This Week's Goals

- Remind ourselves of the concept of energy and energy conservation
- Apply the conservation of energy to a variety of systems
- Develop the mathematical tools to analyze energy conservation in more complex systems
- Connect our new understanding of energy conservation to our previous work on forces and motion

Clicker Question 12-1

Which of the following are true about a point particle? (Use 1/A - True, 2/B - False)

12-1a. A point particle has no size.

12-1b. A point particle can have no mass.

12-1c. A point particle can have no charge.

12-1d. A point particle can have no internal energy.

Clicker Question 12-2

Einstein's proposed total energy for a particle of mass m moving at speed v is given by $E = \gamma mc^2$, where $\gamma = 1/\sqrt{1 - v^2/c^2}$. We take the limit as $v/c \to 0$ to find the total energy of a particle at rest. Which terms below appear in the Taylor expansion of γ in powers of v/c?

a	b	c	d
1	v/c	$(v/c)^2$	$(v/c)^3$

1.) a only 2.) a and b 3.) a and c

4.) b and d 5.) all terms

Clicker Question 12-3

Which of the following are statements of the conservation of energy?

1. The total energy of a system is constant.
2. $\Delta E_{system} = 0$
3. $\Delta E_{system} = W + Q$
4. $\frac{dE_{system}}{dt} = 0$
5. All of the above